

RoboGuardian

RoboGuardian:
Governed execution
is **more resilient** than
open-loop execution
when the world changes.



github.com/rootz-synapsa/roboguardian



WORLD STATE MONITOR



ENVIRONMENT CHANGE DETECTED

OBJECT SHIFT
LIGHTING CHANGE



GOVERNED EXECUTION



OPEN-LOOP EXECUTION



The world changes. The plan doesn't.



100% failure
in 10 deterministic
disturbance trials

tested object-displacement condition



TEST CONDITION

A deterministic lateral displacement is applied to the target object during closed-loop execution.



OUTCOME

The robot fails to recover and does not successfully complete the task in any trial.

RESULT OVERVIEW

Success Rate

0%
0 / 10



100% FAILURE IN ALL TRIALS

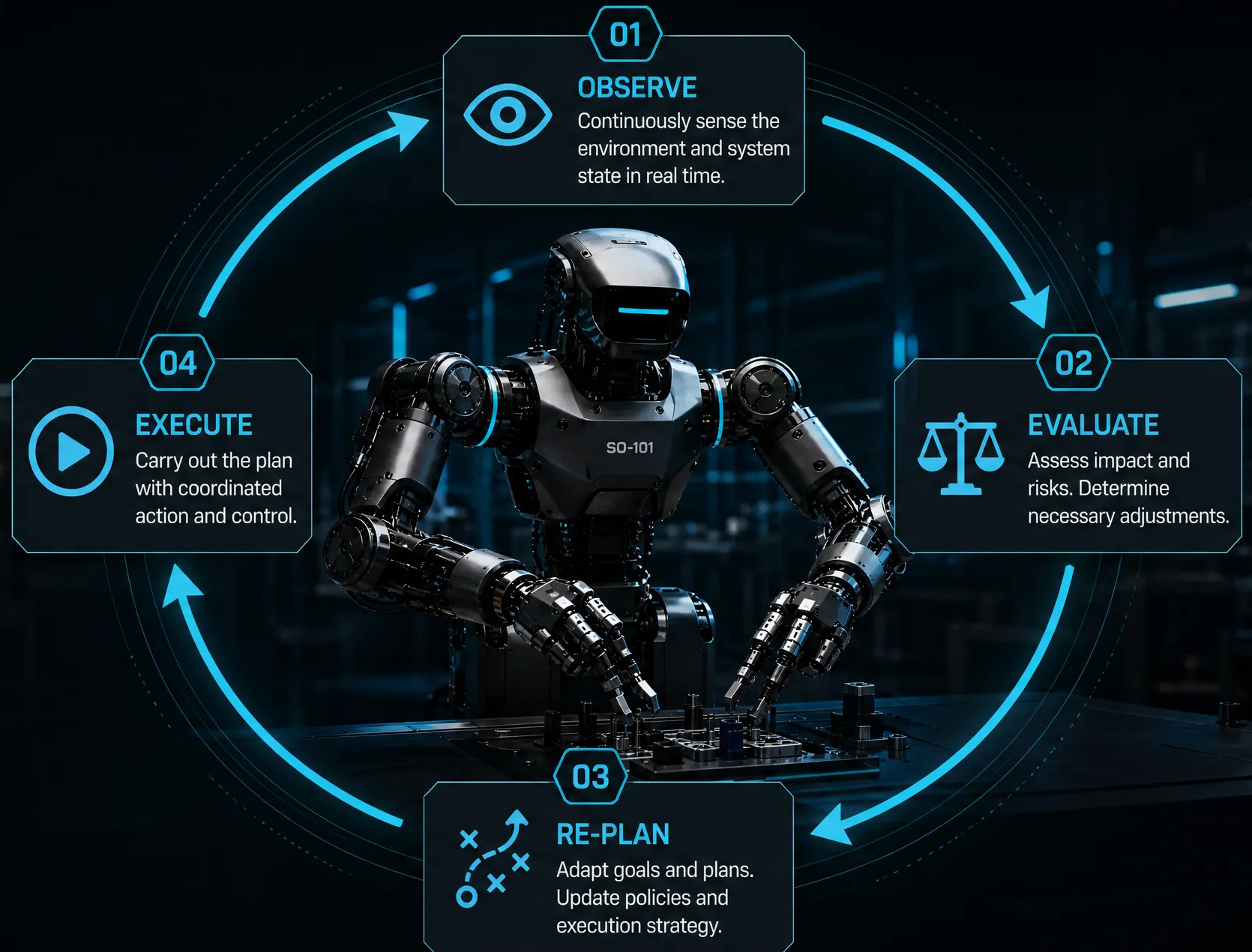


The Approach

RoboGuardian uses a continuous governance loop to **sense, assess, adapt,** and **act**—keeping execution aligned as the world changes.

THE GOVERNANCE LOOP

A closed-loop system that continuously adapts plans and execution in response to environmental change.



Section 18 Guarantee

</>

```
01 def recover(self, stale_target, ...):  
02     trace = Trace(stale_target) # Log only  
03  
04 ► self.execute(fresh_target)
```



STALE TARGET

Logged for traceability
(Never executed)



The stale plan is
NEVER executed



FRESH TARGET

Executed for safety
and correctness



Experiment Design



Apples-to-apples
comparison:

3 Arms, 10 trials each,
identical seeds & disturbances



10 TRIALS
EACH ARM



IDENTICAL
SEEDS



IDENTICAL
DISTURBANCES

DETERMINISTIC TRIALS

ARM A



ARM B



ARM C



SLIDE 6

The Results



Head-to-head reliability:
Zero stale actions executed
by RoboGuardian.

ARM	STALE ACTIONS EXECUTED / 10 TRIALS
Arm A	✓✓✓✓✓✓✓✓✓✓ 10/10
Arm B	✗✗✗✗✗✗✗✗✗✗ 0/10
Arm C	✓✓✓✓✓✓✓✓✓✓ 10/10

RELIABILITY OVERVIEW



RoboGuardian
executed zero stale
actions across all
trials.

■ Arm A	10/10
■ Arm B	0/10
■ Arm C	10/10

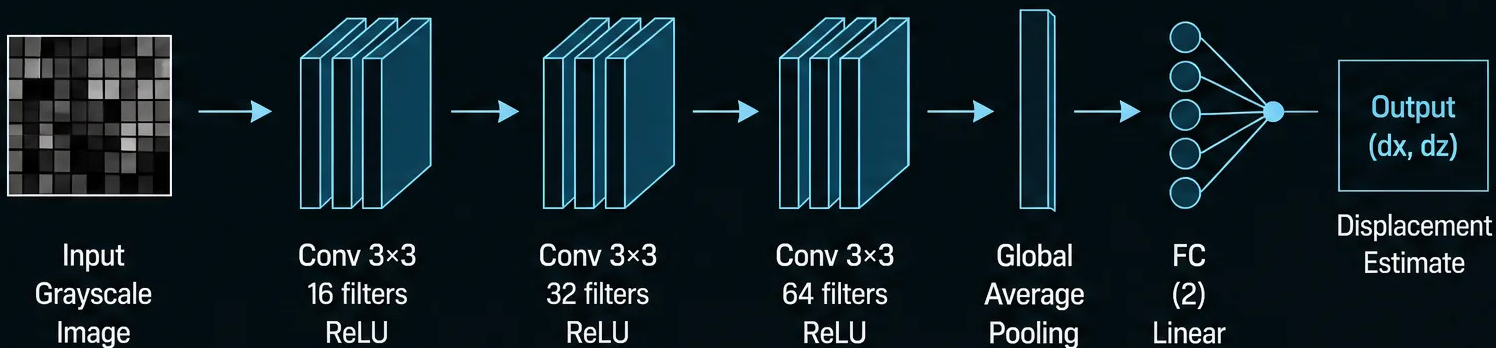


Perception without ground truth

A tiny CNN **learns reliable displacement** estimates without labeled data.



TINY CNN PIPELINE



~146K
parameters



~0.7 cm
mean accuracy



OpenVINO: 0.44x
vs ONNX Runtime



<2 ms
CPU inference latency

Reproducible evidence artifacts

Every gate emits JSON evidence and event traces.

EVIDENCE CHAIN

- | | | | | | |
|----|---|-------------|--------|-----------------------------|---|
| 01 |  | Environment | -----> | results/g1_environment.json |  |
| 02 |  | Baseline | -----> | results/g2_baseline.json |  |
| 03 |  | Failure | -----> | results/g3_failure.json |  |
| 04 |  | Detection | -----> | results/g4_detection.json |  |
| 05 |  | Recovery | -----> | results/g5_recovery.json |  |
| 06 |  | Reliability | -----> | results/g6_reliability.json |  |

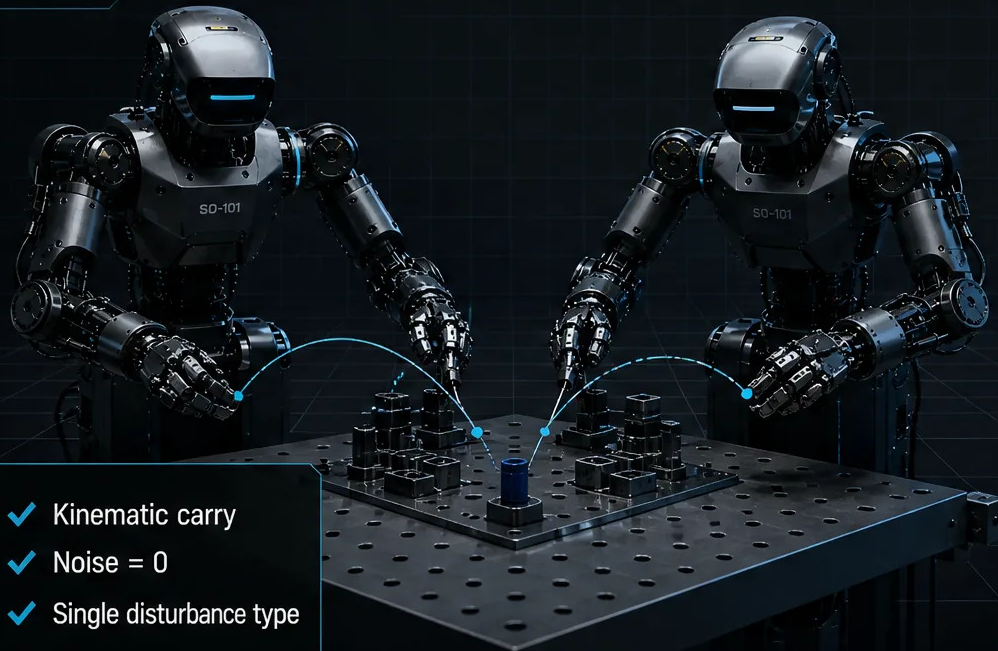


Honest Limitations



Honest Limitations:
Kinematic carry, noise=0, single disturbance type

SIMULATION (IDEALIZED)



- ✓ Kinematic carry
- ✓ Noise = 0
- ✓ Single disturbance type

VS

REALITY (BOUNDARIES)



- ⚠ Model mismatch
- ⚠ Sensor & actuation noise
- ⚠ Unmodeled dynamics
- ⚠ Environment variability
- ⚠ Multi-type disturbances



Our results are promising—but scoped. Generalization to broader disturbances and real-world noise remains future work.

Resilience > Speed.



Scan to
clone & run.



github.com/rootz-synapsa/roboguardian